

Mahi Pasarkar

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Full-stack software engineer building cloud infrastructure tooling in Go, Python, and React. Designs and leads implementation of systems that inform million-dollar decisions at scale. Published first-author researcher (NASA/IEEE).

Education

Rensselaer Polytechnic Institute

Aug 2020 - May 2024

Bachelor of Science in Computer Science (AI and Data Concentration) and Games and Simulations (Dual Major) GPA: 3.7

Research

Natural Language Processing for Structuring Disease Data | First Author

Jan 2024 - Dec 2024

<https://ntrs.nasa.gov/citations/20240015043>

- First author, published by NASA and IEEE. Used natural language processing methods to transform unstructured disease reports into machine readable format. Published parsing system using open science principles to aid research.
- Team Lead, designed system architecture in Python and divided work among coauthors to meet weekly OKRs.
- Gained domain knowledge by coordinating with experts in epidemiology, meteorological data, and disease modeling.

Experience

GEICO | Full Stack Software Engineer II

Jul 2024 - Present

- Informed million dollar hardware purchase decisions by designing and leading team to implement Cloud Repatriation Capacity Modeling system that takes services on the public cloud and predicts the resources required if repatriated to on-prem datacenters.
- Tracked millions in cloud spend and utilization by designing and coordinating end to end development of cloud spend dashboard. Designed backend specification, implemented back end and front end UI with custom graph components.
- Resolved several recurring P0 bugs in all our systems by creating design guidelines for data quality, identifying key risk areas per each system, and design patterns to avoid these risks or centralize them so they are easily discoverable.

Submittity | Full Stack Developer

Jan 2023 - Aug 2023

- Developed EdTech software for computer science courses used by universities, using PHP, Python, and PostgreSQL
- Developed major site features such as overhauling calendar system with customization and filters and fixing major bugs such as assignment setting bugs and security for zip downloads that would expose answers to students.

Projects

Yonderia.com | Plan Trips Beautifully!

- Full stack Next.js and Postgres app for planning trips with friends. Has accounts, comments, event planning, custom theming, and interactive map.

Project Biofuel | Cheap Sustainable Protein

- Experiment with growing Wolffia Globosa, abundant plant with 40% protein content. Standard crops have too much manganese and iron for consumption. Designed custom fertilizer based on bioconcentration factor of key nutrients.

Machine Learning Projects

- Implemented supervised learning methods from scratch such as linear model with legendre feature transforms, linear regression, k-nearest neighbors, radial basis functions, neural networks, and support vector machines.
- Created Q-Learning library to train in-game agents using C# and Unity. Created browser demo with live training:
<https://github.com/mahi-pas/Q-Learning-vs-PPO-Machine-Learning>

Skills

Languages: C++, C#, C, Python, Java, PostgreSQL, HTML5, CSS, Javascript, PHP, Go, Rust

Tools/Libraries: Numpy, Pandas, Matplotlib, Unity, Godot, Unreal, ReactJS, NextJS, Nginx, Systemd

Concepts: Machine learning methods, Linear Algebra, Game Design, System Design, Server Management